

Benyu Wang

University of Michigan, Ann Arbor, US

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Interests

- Broad interest in **Theoretical Computer Science**.
- Study and research experience in **Graph Algorithms / Combinatorics**.

Education

University of Michigan

Ann Arbor, US

PhD student at Computer Science and Engineering (CSE)

Aug 2023 – Now

- Advisor: *Prof. Thatchaphol Saranurak*
- GPA: **4.00**/4.00

Tsinghua University (Yao Class)

Beijing, China

B.Eng., Computer Science and Technology

Aug 2019 – Jun 2023

- Advisor: *Prof. Ran Duan*
- GPA: **3.90**/4.00 & For Yao Class courses only: **3.98**/4.00

Selected Courses: (A+/A stand for 95-100 in 100-point values, A+ is the best grade)

- Mathematics for Computer Science (A+) & Mathematics for Artificial Intelligence (A+)
- Theory of Computation (A+) & Design and Analysis of Algorithms (A+)

Experiences

University of Michigan

Ann Arbor, US

Undergraduate Research Intern (Visitor)

Feb 2022 – Aug 2022

INSAIT

Sofia, Bulgaria

Research Intern (Visitor)

Sep 2024 – Dec 2024

Teaching

Coach of the UMich ICPC team

2025 – Now

Graduate Student Instructor, Introduction to Algorithms

Winter 2025, University of Michigan

Undergraduate algorithm course instructed by Prof. Thatchaphol Saranurak.

Teaching Assistant, Theory of Computation

Spring 2023, Tsinghua University

Undergraduate complexity theory course instructed by Prof. Ran Duan.

Teaching Assistant, Design and Analysis of Algorithms

Fall 2022, Tsinghua University

Graduate-Level algorithm course instructed by Prof. Ran Duan.

Services

Conference Review: STOC, FOCS, SODA, ICALP, ITCS, ESA, COCOON.

Journal Review: Algorithmica, TALG.

Selected Awards and Scholarships

Finalist from Theory Lab in CSE Honors Competition

2025

ICPC Astana World Finalist (Rank 2 in ICPC ECNA)

2023-2024

Excellent Academic Scholarship & **Excellent Art** Scholarship of Tsinghua University

2022

Gold Medal in Chinese Mathematical Olympiad (CMO) (ranked 86th in China)

2018

Silver Medal in National Olympiad in Informatics (NOI) (ranked 90th in China)

2018

Publications

Tight Conditional Lower Bounds for Vertex Connectivity Problems

Zhiyi Huang, Yaowei Long, Thatchaphol Saranurak, Benyu Wang

◦ STOC 2023, [arXiv:2212.00359](#).

Undirected 3-Fault Replacement Path in Nearly Cubic Time

Shucheng Chi, Ran Duan, Benyu Wang, Tianle Xie

◦ ICALP 2025, [arXiv:2411.18312](#).

Near-Optimal Fault-Tolerant Strong Connectivity Preservers

Gary Hoppenworth, Thatchaphol Saranurak, Benyu Wang

◦ FOCS 2025, [arXiv:2510.02562](#).

Connectivity Oracle Under Vertex Failures by Shortcutting Unbreakable Decomposition

Xizhe Li, Yaowei Long, David Pidugu, Thatchaphol Saranurak, Benyu Wang

◦ ICALP 2026, [arXiv:2605.07168](#).

Approximating Directed Minimum Cut and Arborescence Packing via Directed Expander Hierarchies

Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Benyu Wang

◦ Manuscript, in submission, [arXiv:2512.05300](#).

Strong Connectivity Oracle for Multiple Failures via Rooted Flow-Augmentation

Aditya Anand, Thatchaphol Saranurak, Benyu Wang

◦ Manuscript, in submission